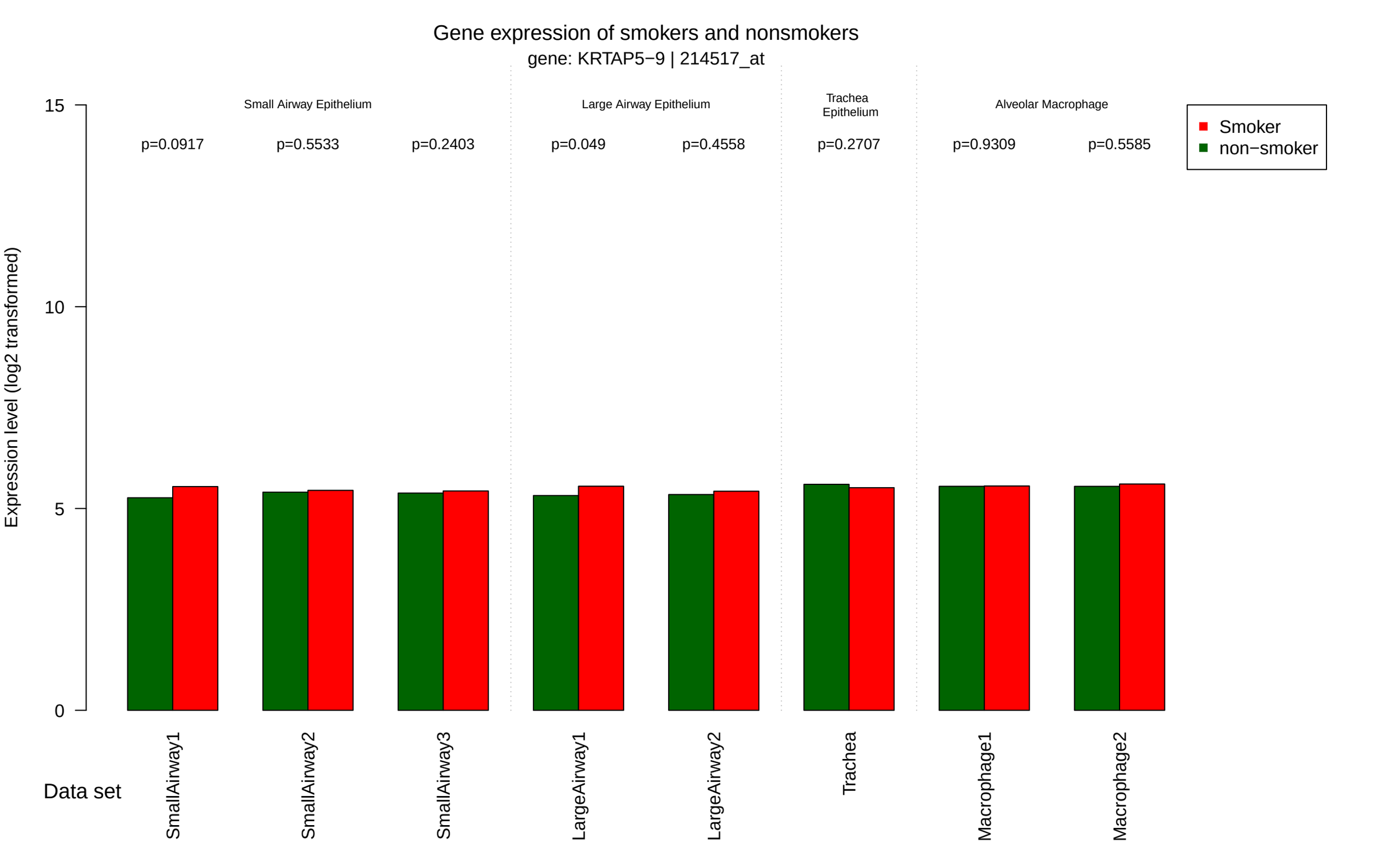
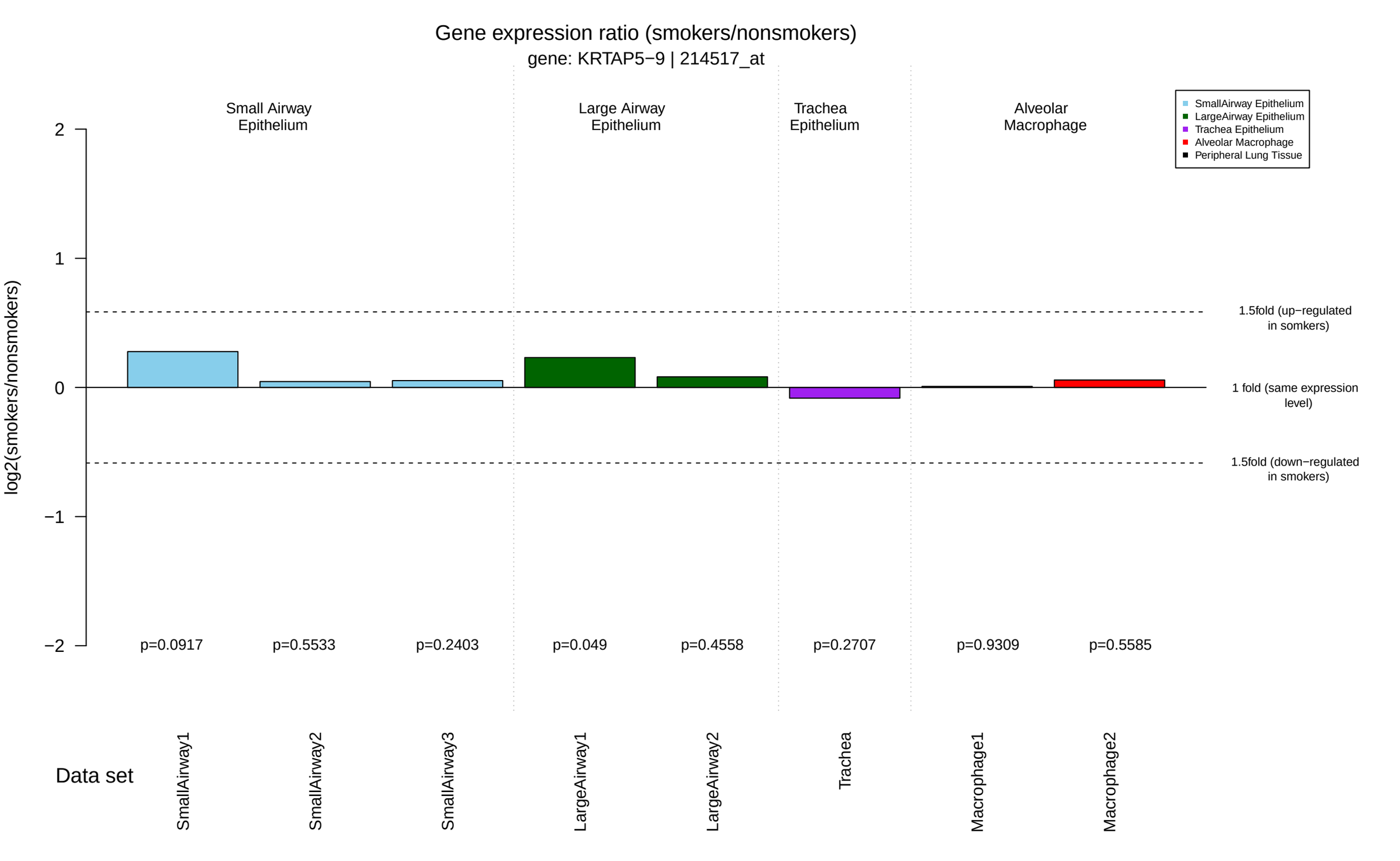
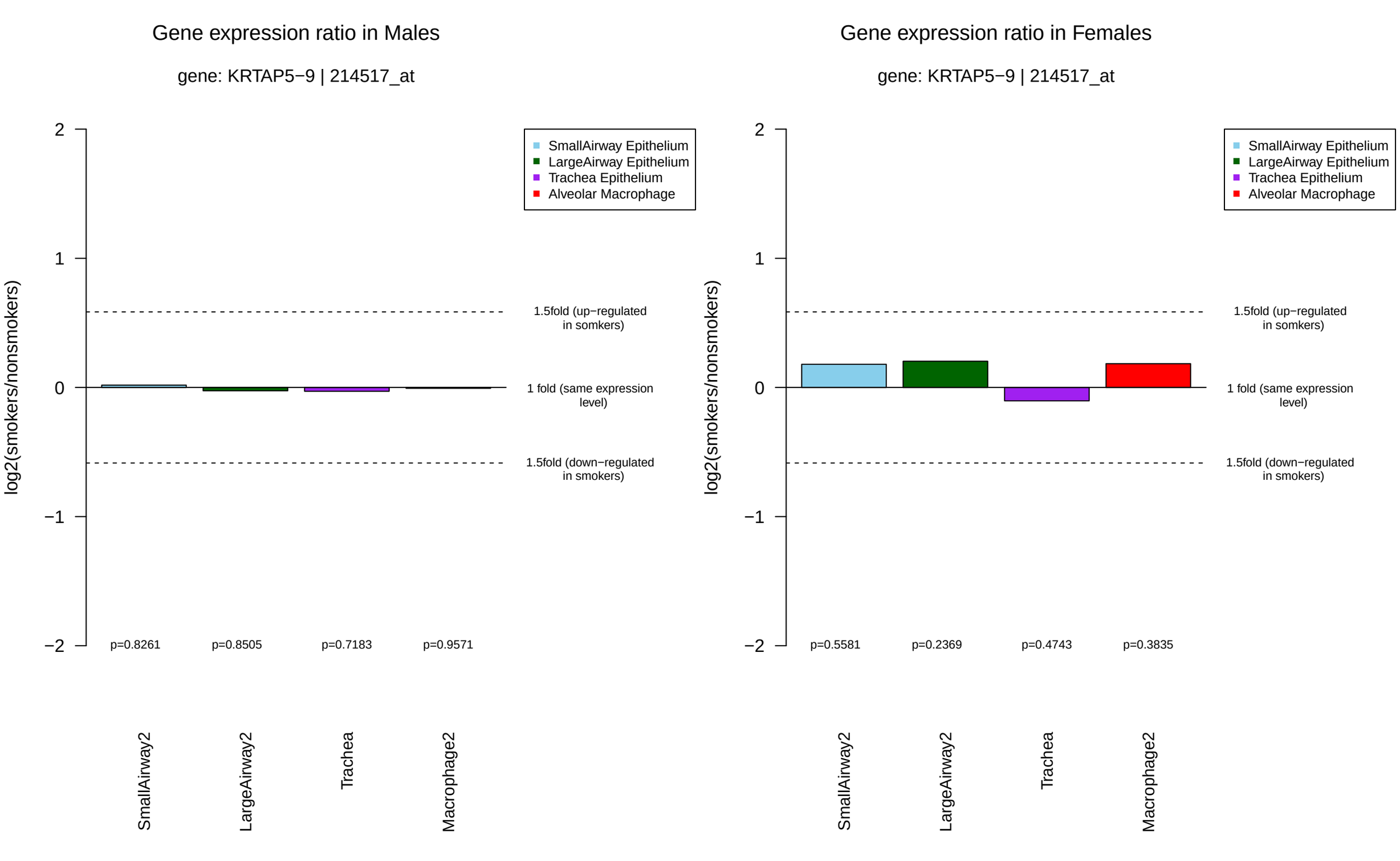
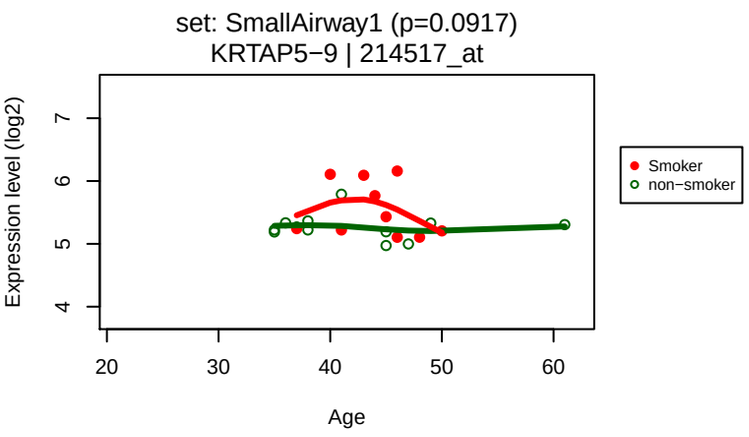
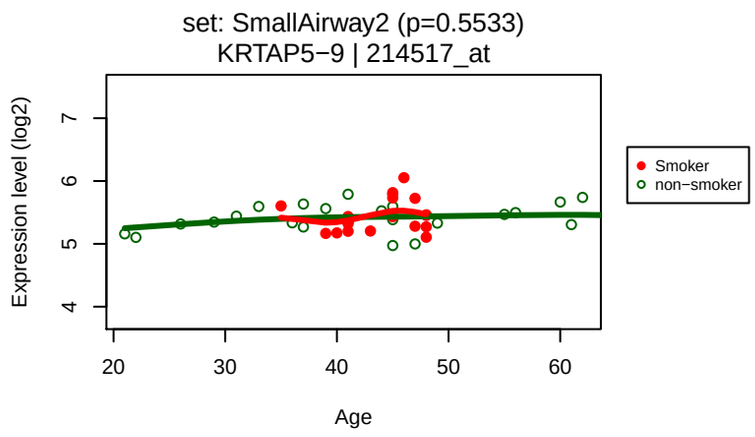
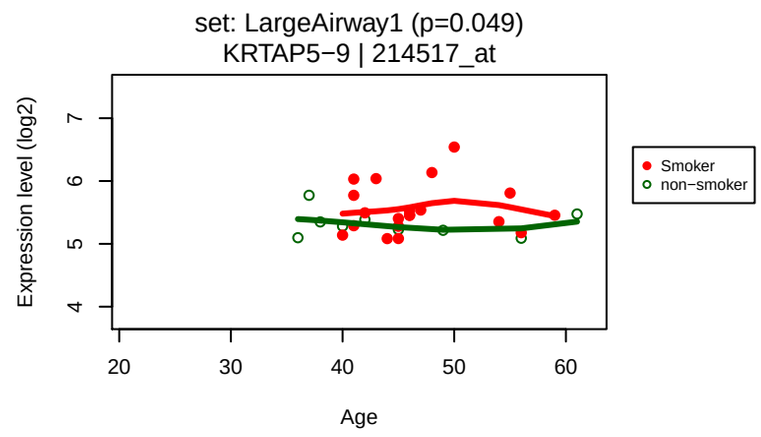

Gene expression of smokers and nonsmokers
gene: KRTAP5−9 | 214517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0917
p=0.5533
p=0.2403
p=0.049
p=0.4558
p=0.2707
p=0.9309
p=0.5585
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

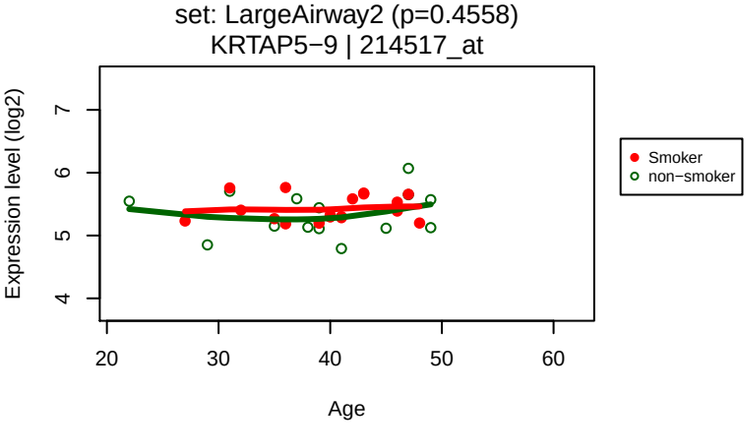

Gene expression ratio (smokers/nonsmokers)
gene: KRTAP5−9 | 214517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0917
p=0.5533
p=0.2403
p=0.049
p=0.4558
p=0.2707
p=0.9309
p=0.5585
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

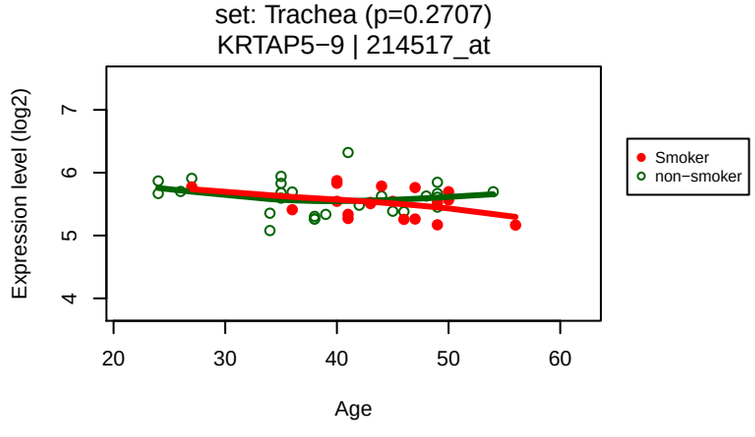

Gene expression ratio in Males
gene: KRTAP5−9 | 214517_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8261
p=0.8505
p=0.7183
p=0.9571
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRTAP5−9 | 214517_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5581
p=0.2369
p=0.4743
p=0.3835
SmallAirway2
LargeAirway2
Trachea
Macrophage2

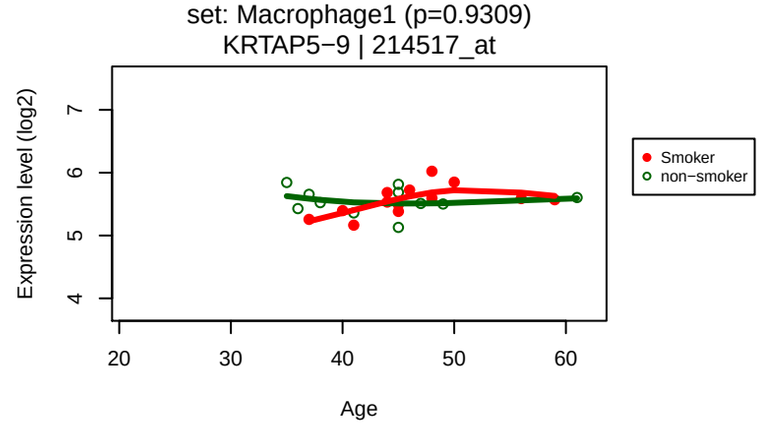

set: SmallAirway1 (p=0.0917)
KRTAP5−9 | 214517_at
Expression level (log2)
Age
Smoker
non−smoker

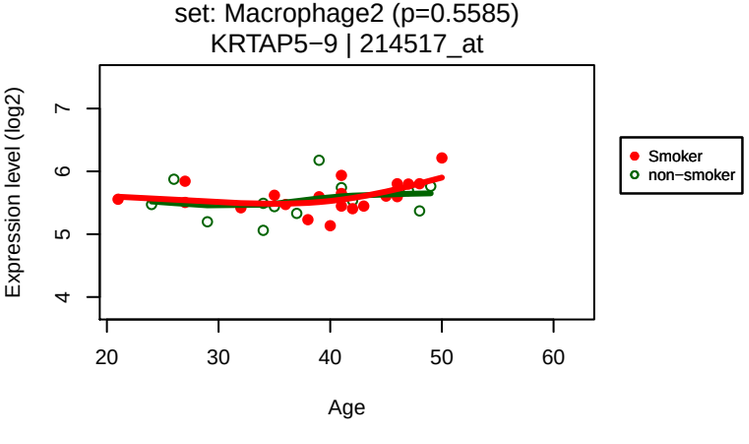

set: SmallAirway2 (p=0.5533)
KRTAP5−9 | 214517_at
Expression level (log2)
Age
Smoker
non−smoker

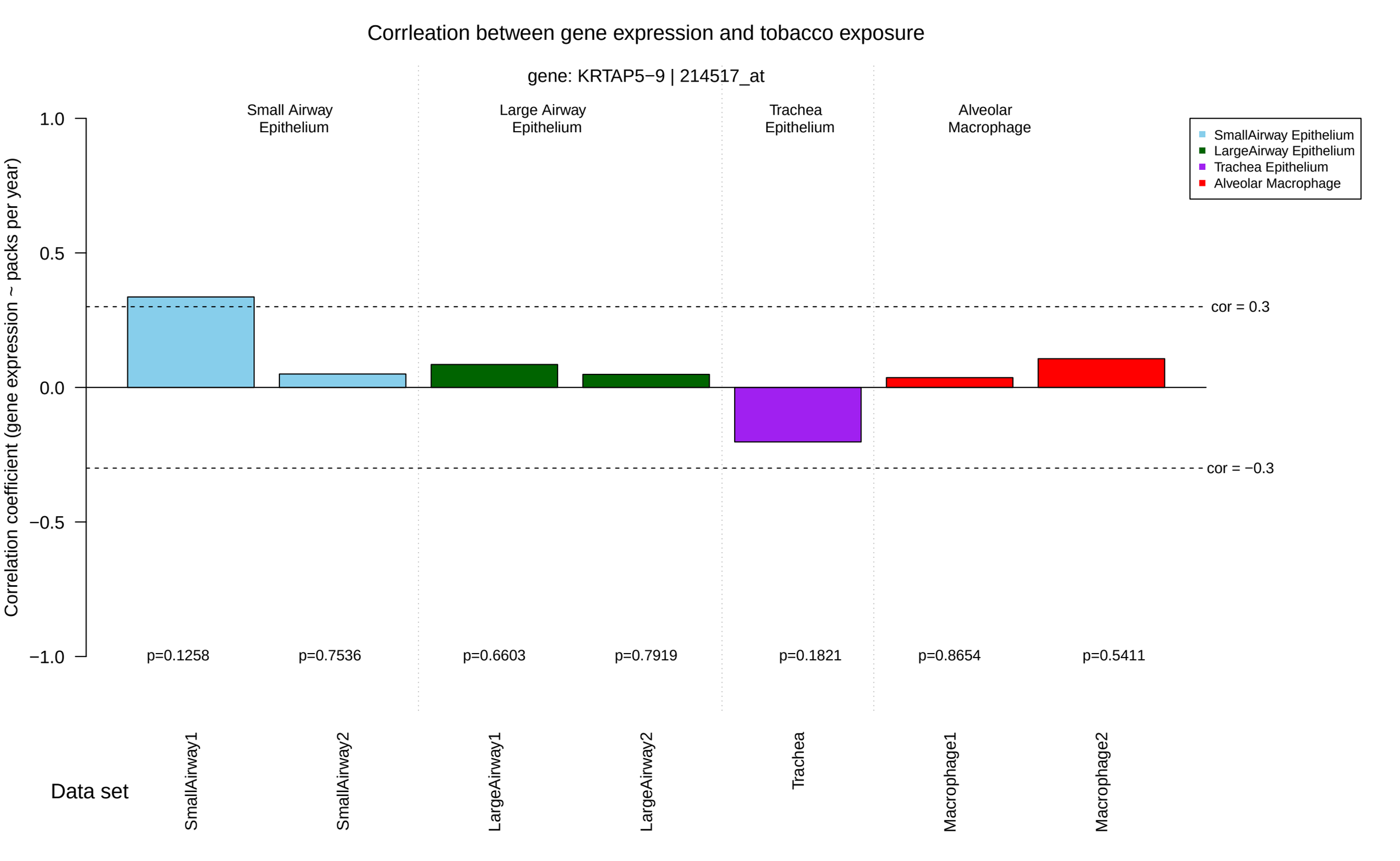

set: LargeAirway1 (p=0.049)
KRTAP5−9 | 214517_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4558)
KRTAP5−9 | 214517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2707)
KRTAP5−9 | 214517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9309)
KRTAP5−9 | 214517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5585)
KRTAP5−9 | 214517_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRTAP5−9 | 214517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1258
p=0.7536
p=0.6603
p=0.7919
p=0.1821
p=0.8654
p=0.5411
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2